MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
MAKE-UP EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**B. TECH**	**Course Type**	**C**
Course Name	**WATER SUPPLY ENGINEERING**	**Course Code**	**CIE 2125**
Semester	**III**	**Max Marks**	**50**
Date		**Duration**	**3 hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Describe population forecasting and elaborate any three methods used.	**4**	CO1	1,2,6,7,12	**L2**
1B	Briefly explain the factors that affect the design period of water supply system.	**4**	CO1	1,2,6,7,12	**L2**
1C	Discuss any two physical characteristics of water in detail.	**2**	CO2	1,6,7,8,12	**L2**
2A	Explain in detail any two operational problems in rapid sand filters.	**4**	CO3	1,2,3,7,12	**L2**
2B	Describe fluoridation and defluoridation process used in water treatment. Explain the Nalgonda technique used for defluoridation.	**4**	CO3	1,2,3,7,12	**L2**
2C	Illustrate breakpoint chlorination with a neat sketch	**2**	CO3	1,2,3,7,12	**L2**
3A	Determine the quantity of alum required to treat 25 MLD of water in a treatment plant where the alum dose is 15 mg/l. Find the amount of carbon dioxide gas liberated during the treatment process. Given the molecular weights as: Al = 27, S = 32, O = 16, H = 1, Ca= 40 and C =12. Assume the following: 1 mole of Alum = 3 moles of $Ca(HCO_3)_2$. 1 mole of Alum = 6 moles of CO_2	**4**	CO4	1,2,3,7,8,12	**L3**
3B	Calculate the amount of pure lime and soda needed to treat 110,000 liters of a raw water sample containing 350 mg/L alkalinity, 100 mg/L hardness as $CaCl_2$, and 90 mg/L hardness as $MgCl_2$. Also, calculate the quantity of slaked lime required if it purity is 80%. Use the following relation: *a) Total Lime requirement = 0.74*(2*temperorary Magnesium hardness + temperorary Calcium hardness + permanent Magnesium hardness)*	**4**	CO4	1,2,3,7,8,12	**L3**

<table>
<tr><td></td><td>b) Total Soda requirement = 1.06*(permanent Calcium hardness + permanent Magnesium hardness)</td><td></td><td></td><td></td><td></td></tr>
<tr><td>3C</td><td>Calculate the dosage and chlorine demand at a water treatment plant designed for 5,600 person, with water demand of 270 liters per capata per day. Assume peak factor as 2.5 and daily chlorine requirement as 7 kg/day. The residual chlorine after a 10-minute contact time is 0.13 mg/L.</td><td>2</td><td>CO4</td><td>1,2,3,7,8,12</td><td>L3</td></tr>
<tr><td>4A</td><td>A pump with a flow rate of 2,000,000 liters per hour is required to transport water over a 2,100 m distance. The water is pumped from a canal at an elevation of 15 m to a storage reservoir at 50 m elevation. Given a friction factor of f = 0.01, calculate the economical diameter for the rising main, and what is the required water horsepower for the pump.</td><td>5</td><td>CO5</td><td>1,2,6,7,12</td><td>L3</td></tr>
<tr><td>4B</td><td>List various types of water intake strutures. Explain any one of the above in detail.</td><td>3</td><td>CO5</td><td>1,2,6,7,12</td><td>L2</td></tr>
<tr><td>4C</td><td>Explain measures to prevent water leakage in a water distribution system.</td><td>2</td><td>CO5</td><td>1,2,6,7,12</td><td>L2</td></tr>
<tr><td>5A</td><td>Using Graphical method calculate the storage required to supply the demand shown in the following table. The pumping is to be carried out at a constant rate from 8 AM to 4 PM.
<table>
<tr><td>Time (hours)</td><td>0 - 4</td><td>4- 8</td><td>8- 12</td><td>12 - 16</td><td>16 - 20</td><td>20 - 24</td></tr>
<tr><td>Demand (MLD)</td><td>0.35</td><td>0.9</td><td>2.5</td><td>1.5</td><td>1</td><td>0.5</td></tr>
</table></td><td>5</td><td>CO5</td><td>1,2,6,7,12</td><td>L4</td></tr>
<tr><td>5B</td><td>Explain the factors governing the selection of pipe material used in water supply system.</td><td>3</td><td>CO5</td><td>1,2,6,7,12</td><td>L2</td></tr>
<tr><td>5C</td><td>Describe pipe appurtenances and explain any one appurtenance in detail</td><td>2</td><td>CO5</td><td>1,2,6,7,12</td><td>L2</td></tr>
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MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***08***
2	***02***
3	***10***
4	***10***
5	***20***

BL	Total Marks
2	30
3	15
4	5
5	-